

IT-OT Integration Challenges in Utilities

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Abstract— Emerging industrial IoT presupposes integration of operational technology (OT) networks with information technology (IT) networks which hitherto were isolated. IT networks focus primarily on PAIN (Privacy/Confidentiality, Authentication, Integrity and Non-repudiation) OT networks focus primarily on AIC (Availability, Integrity and Confidentiality). The conflicting requirements of availability and confidentiality is a challenge that needs to be overcome in a balanced way to reap the benefits of IT-OT integration. This balancing act requires understanding of the OT domain. This paper considers electrical utility as the use case and brings out the considerations in achieving an acceptable IT-OT framework.

Keywords- Information Technology, Operation Technology, IT/OT convergence, IoT, Electric Utility, SCADA, OMS.

I. INTRODUCTION

Today's need is to make information available at each door for better operation, decision making and future prediction. Information Technology (IT) / Operation Technology (OT) integration can make this possible.

OT convergence with IT is integration of IT systems, used for data centric computing with OT systems, used to monitor events, processes and devices and make adjustments in enterprise and associated industrial operation. Information Technology includes any use of computers, storage, networking and other physical devices, processes to create, process, store, secure and exchange all forms of electronic data. Whereas OT is associated specific functions like manufacturing and industrial environments, which include industrial control systems such as supervisory control and data acquisition (SCADA).

While IT inherently covers communications as a part of its scope, OT is isolated with networked technology. Many devices for monitoring or adjustment were not computerized and those with compute resources generally use proprietary protocols and programmable logic controllers (PLC) rather than technologies that afford full computer control.

With Technological improvement, devices like wireless sensors and actuators are being integrated in industrial environments such as Paper industry, Electric Utilities, Oil Refineries and other factories. The integration of automation, communications and networking in industrial environments is an integral part of the growing Internet of Things (IOTs).

Monitoring is simplified by using IOT, where physical devices are monitored and controlled remotely through Internet. IOT is used in many of industries now, especially for security and in hazardous locations where human presence can be minimized. IOT made data available to operators, management, stakeholders and consumers wherever it is required. This increases transparency.

Most of reputed global companies are in process of integrating OT with IT, in this paper, we will discuss about Electrical Distribution Utility combining these two domains. In making of this process which is new will have different challenges during implementation and advantages after successful implementation. These will be shared in this paper. Further paper is organized as follows: Section II gives idea of Information Technology (IT) and Operation Technology (OT). Section III elaborates about need of IT/OT convergence. Section IV. Lists out advantages, challenges and solutions. Section V. is case study in electrical utility where in OMS is implemented by integrating IT with OT. Finally Section VI concludes the paper.

II. INFORMATION TECHNOLOGY(IT) & OPERATION TECHNOLOGY (OT)

Information Technology (IT) –

Any electronic device that can store (eg. hard drives, tapes), calculate (eg. Software applications), transport (eg. Communication network, data formats) Data will come under umbrella of Information Technology (IT).

With increasing data, IT emerged with technology which made it available to every individual fulfilling each need. Databases were used for information storage and for retrieve information when required.

Operation Technology (OT) -

Operation Technology (OT) is combination of hardware and software that monitors and controls performance of physical devices.

OT was primarily used in Industries for manufacturing, transportation, utilities etc. Many of these devices used for monitoring and controlling physical devices were mechanical and few devices which can communicate have proprietary protocols and incompatible with any other formats.

This led to monopoly in the industry and unrealistic prices. Later standards and open protocols such as Modbus[4], IEC 101[2], DNP[3] and other protocols made a start for standardization, presently in electrical substations, IEC61850[1] is a standard which gives guidelines for all devices and make easy understanding between any devices irrespective of make and model.

Figure 1. gives details of different applications in IT and OT with phase wise integration.

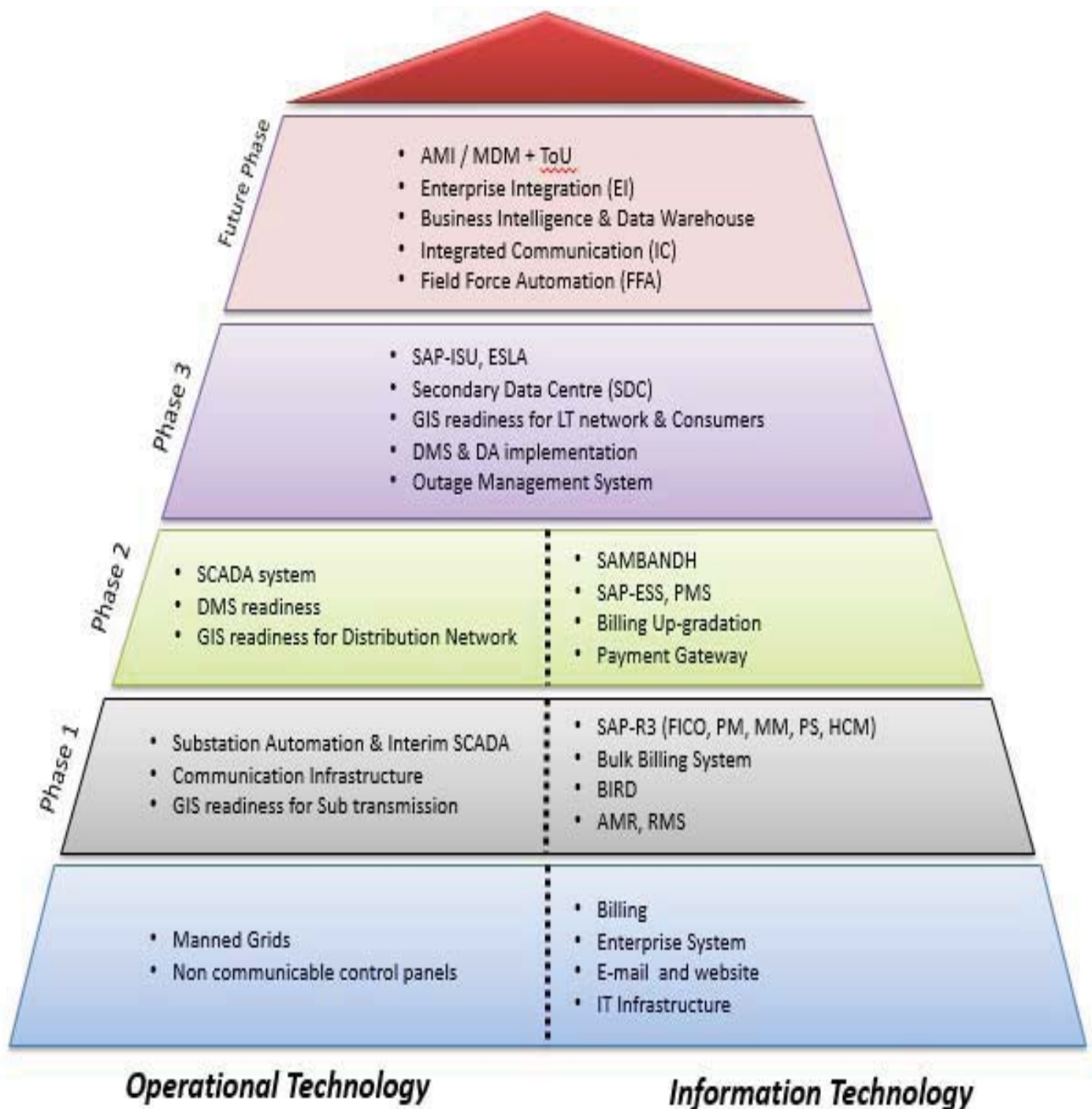


Fig. 1 : IT/OT Convergence phase wise in Utility[5]

III. NEED FOR OT INTEGRATION WITH IT

Improvement in customer satisfaction, error free implementation of processes/calculations, transparency and reduction in time of process/manufacturing/calculating were always key areas of interest for every business. Technology can be used as a buddy for any business to progress.

Operation technology alone can monitor devices from specified location where operators are located. If OT can be accessed from different locations and by management, thus simplifies decision making and reduces transportation for any organization. This can be achieved by integration with Information Technology.

As physical devices are becoming advanced, IT/OT convergence can be seen in many of the industries. Wireless connectivity has provided operational technology with better monitoring systems and the ability to control physical devices remotely. Advances in machine-to-machine communication and machine learning have created a difference, allowing the data that physical devices produce to be analysed in real time in order to facilitate autonomy, allow preventive maintenance and improve up-time of the devices.

Internet has moved into industrial control systems and supervisory control and data acquisitions (SCADA) networks. However, operations technology faces the same malware, identity management and access control security challenges that IT faces. The difference is that vulnerabilities in an OT system can leave critical infrastructure at risk of sabotage that can result in life or death situations if not addressed.

Figure 2. gives overview of utilization of IT/OT integration

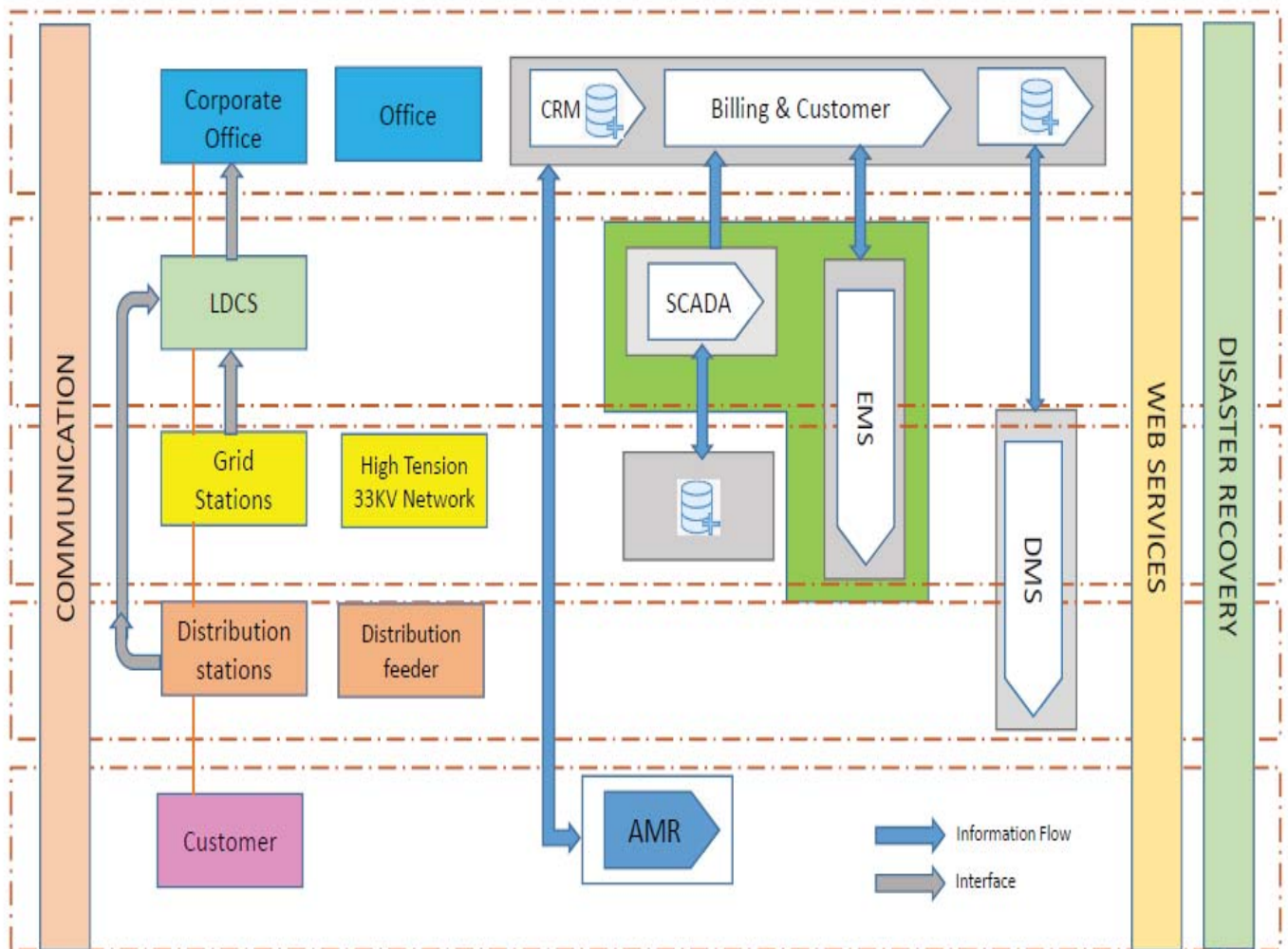


Fig. 2 : Utilization of IT/OT Integration in Utility[5]

In the above figure, it is clearly visible that Grid Substations (SCADA, OT), EMS (SCADA, OT), Distribution stations (DMS, OT & CRM (SAP, IT) are interconnected and share information with each other enabling an efficient and smooth operation.

IV. ADVANTAGES, CHALLENGES AND SOLUTIONS

Advantages-

IT/OT convergence enables more direct control and complete monitoring, with easier analysis of data from these complex systems from anywhere in the world. Few of advantages are mentioned below.

Digitalization- Every information will be available in digital form.

Asset mapping/management- Assets are mapped making ease for maintenance planning and to monitor age of asset. Average increase in age of asset will be around 20%.

Centralization- Decision making from central location.

Productivity improvement- minimization of breakdown due to prediction, which leads to high productivity.

Data accuracy at all levels- Information from physical devices reaches without any human intervention thus reducing errors and providing accurate data.

Transparency- Data availability at different hierarchies in organization gives transparency.

Quick and correct Decision making- Quick decision can be taken such as load scheduling & load shedding by monitoring load.

SAP/GIS(Geographic Information System) integration with OT- enterprise solutions such as SAP and GIS can be integrated with SCADA/DMS.

Better consumer service- Customer service executive using Customer Relationship Management (CRM) module in SAP will have access to SCADA and GIS, can provide correct reason for outage and estimated time. Outage time can be reduced to 30 minutes, before integration it used to be around 2 hours.

Automated Meter Reading (AMR) integrated with Billing module in SAP enabling error free reading which adds to customer delight. Billing efficiency will be increased to more than 99% and revenue collections for important/high revenue consumers will be increased.

Challenges-

There has been an increasing trend of companies expanding their use of Operational Technology (OT) beyond traditional manufacturing and logistics applications into areas of the business that traditional IT previously supported. With OT devices and embedded sensors providing increased monitoring and control capabilities, it is not surprising that companies are finding new opportunities to enable business process, facilities and workforce automation using this technology – bringing them closer to the vision of a digital enterprise.

The challenge for CIOs and their organizations is determining to what extent OT should be integrated with traditional IT systems, and then articulating the value case for investing in the integration effort.

OT's modernization through IT integration brings with it the required consideration of security. Many operational

technology systems were never designed for remote accessibility and, as a result, the risks of connectivity were not considered. Such systems may not be regularly updated. The vulnerabilities of OT systems can leave organizations and critical infrastructure at risk of industrial espionage and sabotage. Apart from security, interface with existing IT will be one major challenge as many of OT systems are communicating based on proprietary or non- standard protocols. Few more challenges are mentioned below.

- Access of OT devices on public network leads to wrong operation of companies equipment, misuse of information from devices and mislead organization with wrongly fed data.
- Bandwidth requirements of IT are different from OT requirements, Unlike IT, OT data can have sudden surge due to several reasons which leads to many changes in existing IT system requirements.
- Data from multiple interfaces and integrations between different applications and data formats.

Solutions-

Operational Technology is expected to continue to assume an expanding role in most organizations, enabling greater integration between business processes, physical activities and supporting technology. Integrating OT into the broader IT environment can seem challenging, but the potential benefits are significant. Data Quality Management solutions can make the task of IT/OT integration easier, and enable any company to begin reaping the benefits sooner.

Any company/Organization planning to integrate OT with IT should consider below mentioned points to overcome challenges.

- Proprietary protocols used in OT systems should be standardized. Protocols such as LON (ABB), SPA (ABB) etc. used in Grid Sub Stations (GSS) were replaced with IEC61850 [1] and IEC104 as communication protocol between SCADA at main control centre and Remote terminal Unit (RTU) at GSS.
- OT data should be available in formats accessible by IT. Extensible Markup Language (XML) [6] is used as interface between OT and IT systems
- Interface servers should be defined to make process easy while transferring data.
- Use of Cyber security and to stop unwanted access from hackers.[7]
- Use of Firewall between Enterprise and OT network.
- Policies to be established for all IT users to isolate OT information from IT.
- Data limitation should be introduced which can handle sudden surge.
- IP whitelisting for all OT devices to stop intervention of unwanted access.
- Organization should have own Network, required for faster access of information. In TATA Power-DDL, initially leased line was used for communication which was gradually converted into own network.
- Pre-approved processes are required and timely updating as and when required.[8]

V. CASE STUDY (INDUSTRY: ELECTRICAL UTILITY)

Introduction:

Electric Utility business is to purchase power from generation/transmission companies and supply to consumer.

Brief tasks are

- Purchase of power from Generation units.
- Maintenance and operation of local sub-stations.
- Local transmission to client which have maintenance and operation of several devices (Distribution).
- Metering.
- Billing.
- Revenue collection.
- Asset management.
- Handling consumer complaints.

Apart from these core functions, as a business unit it has other functions like Material handling, human resources, finance etc.

For performing all the above said activities, Utility was dependent on complete man power which lead to high AT&C losses, poor quality of services which gives low consumer satisfaction. Utility used several OT and IT technologies to overcome these issues. Some of technologies used are

- SCADA (OT, for monitoring substations).
- GIS (subset of OT, for asset identification).
- DMS (OT, for monitoring distribution feeders).
- SAP (IT, for Finance, Material management, Human resource management and other business functions)

Earlier OT and IT technologies were used independently, during this phase many processes are put in place and maintenance was made easy but access to OT data was always limited to operators and respective departments. Information was not available easily for management to take a business decisions. Here is necessity for converging OT with IT where in OT applications/ devices made available to required personnel in organization for data analytics. Few of examples are mentioned below.

- Manual billing can be replaced with AMR (Automatic Meter Reading), where consumer meter (OT) will communicate with server (IT) and bill can be generated from utility office and the same can be viewed for management. This reduces manual errors in bill reading and transportation.
- SCADA (OT) information made available at servers (IT), this data can be used for energy audits, asset management and Outage management system.

Traditional Outage Management System-

Consumer call to call centre/ zone office/ field personal and call centre team will forward the same to respective zone office.

Consumer have access to many options and which creates chaos and no centralized location for handling all complaints. Call centre operator do not have all necessary information at his desk to address immediately, making it tough to handle consumer and provide accurate information to consumer.

Integrated Outage Management System (OMS):

Integrated OMS is example of OT convergence with IT. OMS:

An Outage Management System (OMS) is a computer system application used by operators of electric distribution company to assist in process of managing power outages.

An Outage Management System facilitates efficient management of outages by utilizing user configurable logics supported by an geo referenced integrated environment of all systems and processes responsible for handling power supply in the utility.

When Electricity power supply is not available at consumer place, consumer will raise a complaint and utility will restore supply to the consumer.

Utility's responsibility is to restore power as early as possible. Factors that delay are

- Consumer care operator is not aware of cause for breakdown.
- Whole network is not visible to operator or concerned team.
- Locating consumer premises.
- Locating the fault.

An Integrated Outage Management System is integration of call centre application with SAP, SCADA, GIS and AMR/AMI, all essential information is available in the application making process easy. Figure 3. Explains how different applications are linked with each other.

Results after Implementation of OMS and IT OT integration.

- Outage handling time decreased from 2 hours to 30 minutes.
- Count of repeated complaints reduced more than 20% due to grouping of complaints and informing consumers regarding planned outages through SMS.
- Instant information to consumers via SMS regarding interruptions exceeding 30 minutes also reduces complaint count.
- Delta updates in GIS can be incorporated within 1 hour, earlier it used to take a day for manual incremental update / changes.
- Energy Audit application in IT system, which uses Automatic Meter Reading (AMR) will have meter data from SCADA system, can be used in case of no availability of meter or network in AMR.
- Network monitoring tool which monitors communication elements of OT systems reduces break down time and enables utility to take preventive action where in redundant systems are available. This ensures low count for break down cases.

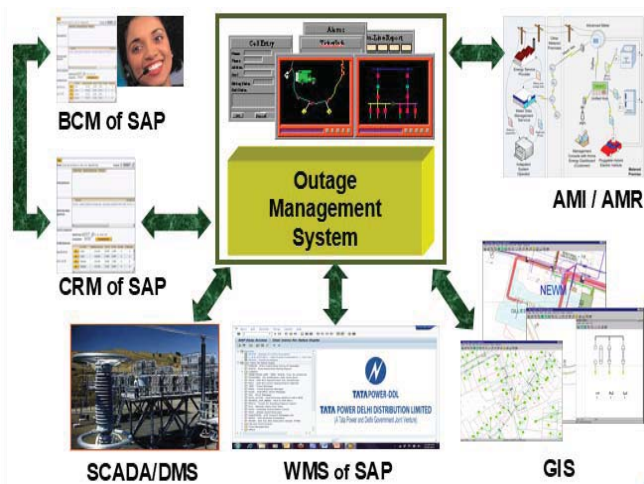


Fig. 3 : Integrated Outage management System [5]

VI. CONCLUSION

Case study presented in this paper is successful implementation of OMS in TATA Power-DDL. Outcome of this case study is reduced efforts of Utility and higher consumer delight due to quick and accurate response. This is possible only because of integration of OT with IT.

Here SCADA/DMS which is part of OT is being integrated with SAP, part of IT enabling Call center operator to access information of Grid Substations & RMUs which are mapped in GIS which gives exact location enabling operator to send field crew to address problem.

Challenges to be handled post implementation.

Process change in itself is big challenge, apart from this, major challenge is difference in update frequency in IT and OT systems. Operating systems, Databases which are part of IT have frequent updates whereas OT system working on IT platform will be updated at later stages, causing interface issues. Handling these issues will be difficult as Original Equipment Manufacturer (OEM) of OT systems have limitations for updating their products.

Similar model can be implemented in different utilities for better results.

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